

Stochastic-cascade parallel track: modified Hasegawa–Wakatani bridge pilot

Living project note

1 September 2026

1 Purpose

This note records an exploratory parallel track to the canonical finite-dimensional nonmodal theory. Its purpose is to test whether a drift-wave/zonal-flow model can support the sequence

field dynamics $\rightarrow$ zonal/nonzonal split $\rightarrow$ scale process $\rightarrow$ Markov test $\rightarrow$ Kramers–Moyal diagnostics.

It does not choose the final plasma convention for the main project and it does not constitute a converged turbulence calculation.

2 Benchmark equations

The pilot uses the Numata-style modified Hasegawa–Wakatani equations

$$\partial_t \zeta + \{\phi, \zeta\} = \alpha(\tilde{\phi} - \tilde{n}) - D_\zeta \nabla^4 \zeta, \quad (1)$$

$$\partial_t n + \{\phi, n\} = \alpha(\tilde{\phi} - \tilde{n}) - \kappa \partial_y \phi - D_n \nabla^4 n, \quad (2)$$

with

$$\zeta = \nabla_\perp^2 \phi, \quad \tilde{f} = f - \langle f \rangle_y.$$

The defining modification is that the parallel/resistive coupling is absent for the zonal modes $k_y = 0$.

The reference numerical run uses a 32^2 periodic grid with $L_x = L_y = 40$, $\alpha = \kappa = 1$, $D_\zeta = D_n = 0.02$, a pseudo-spectral Poisson bracket, two-thirds dealiasing, and fourth-order Runge–Kutta with $\Delta t = 0.05$. The initial perturbations are random with amplitude 10^{-3} and seed 3.

3 Zonal and shell diagnostics

The $E \times B$ kinetic energy is partitioned as

$$E_K = E_{ZF} + E_{NZ}, \quad f_{ZF} = \frac{E_{ZF}}{E_{ZF} + E_{NZ}}.$$

For the analysis window $120 \leq t \leq 310$, the median split is

$$f_{ZF, \text{median}} = 0.84656.$$

The low-ZF ensemble has mean $f_{ZF} = 0.61492$ (191 snapshots), while the high-ZF ensemble has mean $f_{ZF} = 0.90633$ (190 snapshots).

The nonzonal kinetic energy is strongly reduced in the high-ZF ensemble. For representative radial Fourier shells:

$ k $ shell	low ZF	high ZF	high/low
0.0–0.4	0.002388	0.001234	0.517
0.4–0.8	0.024953	0.013018	0.522
0.8–1.2	0.072721	0.019795	0.272
1.2–1.6	0.034627	0.010959	0.316
1.6–2.0	0.010620	0.003630	0.342
2.0–2.6	0.000446	0.000101	0.225

4 Scale process and Markov diagnostics

Zonal modes are removed from ϕ , and increments of the nonzonal potential are formed at separations

$$r/\Delta x \in \{8, 4, 2, 1\}, \quad \Delta s = \ln 2,$$

with the process directed from large to small separation. The present pilot pools x - and y -directed increments; later plasma-quality work must keep anisotropy explicit.

A Gaussian conditional-mutual-information proxy gives

triplet	low ZF	high ZF
8–4–2	0.05034	0.01010
4–2–1	0.21144	0.20541

where values are in nats and zero is the Gaussian-Markov limit.

A discrete Chapman–Kolmogorov test compares the direct transition matrix with the composition

$$P(X_2 | X_0) \stackrel{?}{=} P(X_2 | X_1)P(X_1 | X_0).$$

The occupancy-weighted total-variation errors are

triplet	low ZF	high ZF
8–4–2	0.09387	0.06119
4–2–1	0.08364	0.08688

Thus the larger tested triplet becomes more nearly Markovian in the high-ZF ensemble, whereas the smaller triplet does not.

5 Finite-step drift and diffusion

For the scale step $4\Delta x \rightarrow 2\Delta x$, increments are normalized by the same low-ZF reference standard deviation. Conditional moments are fitted as

$$D^{(1)}(u) \approx a_1 u + a_3 u^3, \quad D^{(2)}(u) \approx b_0 + b_2 u^2.$$

The resulting finite-step coefficients are

coefficient	low ZF	high ZF
a_1	-0.72497	-0.72289
a_3	0.00337	0.00207
b_0	0.34951	0.11664
b_2	0.19309	0.17742

The main pilot signal is an approximately threefold reduction of the diffusion intercept b_0 , while the fitted drift changes very little. This is compatible with suppression of stochastic nonzonal fluctuations by strong zonal organization, but these finite-scale fits are not yet converged Kramers–Moyal coefficients.

6 Current interpretation and gate

The pilot establishes that the proposed observable chain can be implemented in a controlled drift-wave/zonal-flow model and that conditioning on zonal-flow strength changes measurable scale statistics. It does *not* establish causality because the low/high-ZF split is partially confounded with temporal evolution in a single realization.

Before transfer to GENE, the next gate is therefore: independent or stationary ensembles; separate radial/poloidal increments; bootstrap uncertainty; local $D^{(1)}$, $D^{(2)}$, and $D^{(4)}$ at several scale ratios with small- Δs extrapolation; and an original-HW control in which the zonal coupling modification is disabled.

7 References

R. Numata, R. Ball, and R. L. Dewar, *Nonlinear Simulation of Drift Wave Turbulence*, arXiv:physics/0703274 (2007).

A. Hakim, *Solving (Modified) Hasegawa–Wakatani equations*, Simulation Journal JE17 (2026).